Michael Alexander Saxer

B.Sc. Data Science — Electronics Engineer (VET)

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Education

M.Sc. in Computational Science and Engineering, ETH Zürich *09/2026 – 06/2029*
Physics specialization; individual 164 ECTS study plan. Focus on computational modeling, numerical methods, high-performance computing, and physics applications.

B.Sc. in Data Science, Zurich University of Applied Sciences (ZHAW) *09/2022 – 06/2025*
Thesis: *Optimization of Track Clustering for Primary Vertex Reconstruction in Proton-Proton Collisions at the CMS Experiment at CERN. DOI: 10.21256/zhaw-34860*
Specialization in machine learning and computational modeling; additional coursework in PDE numerics and engineering physics.

Federal Vocational Baccalaureate, Engineering/Architecture/Life Sciences *08/2021 – 06/2022*
Final Thesis: *Therapy of Prostate Cancer with Lutetium-177-PSMA-617.*
Bilingual program (English/German).

Research Experience

Undergraduate Researcher, Paul Scherrer Institute (PSI), High Energy Physics Group *08/2024 – 06/2025*
Semester Project and Bachelor Thesis within the CMS Experiment at CERN, supervised by Dr. W. Erdmann (PSI / ETH Zurich). Focused on primary-vertex reconstruction algorithms for the High-Luminosity LHC. As part of this work, I attended the Inverted CERN School of Computing (iCSC 2025) in Meyrin, with hands-on modules in HEP computing, machine learning, and software best practices. Both the semester project (6 ECTS) and bachelor thesis (12 ECTS) were graded 6/6.

- Evaluated the Deterministic Annealing in Blocks algorithm on HL-LHC simulation data and identified block-border effects as the main limitation for reconstruction quality under high-pileup conditions.
- Developed **CascadeDA**, a hybrid vertexing algorithm that combines block-wise pre-clustering with global deterministic annealing.
- Achieved 60% faster clustering than the previous implementation on large-scale CMS tracking datasets while maintaining sub-10 μm z-resolution and a 99% track-to-vertex assignment efficiency.

Work Experience

Teaching Assistant, Zurich University of Applied Sciences - School of Engineering *08/2023 – 06/2024*
In the summer break, I taught mathematics tutorials in the ZHAW Pre-College program (technical vocational baccalaureate). During the semester, I supervised weekly tutorials for first-year engineering students in Calculus I–II and Linear Algebra I–II, and provided additional support in Python programming and study methods.

Apprenticeship in Electronics Engineering (VET Diploma), Siemens Switzerland AG *08/2017 – 08/2021*
Training in electronic circuits, embedded systems, and software engineering. Developed projects in C, C++, and C# and implemented building automation solutions.

Publications and Presentations

L. Stuhlmann, M. A. Saxer, J. Fürst, "*Efficient and Reproducible Biomedical Question Answering Using Retrieval Augmented Generation*," *IEEE Swiss Conference on Data Science (SDS)*, Zurich, 2025. DOI: 10.1109/SDS66131.2025.00029

Presentation: *Quality Assessment of Parallelized Primary Vertex Reconstruction at the CMS Experiment*, Tracker POG Meeting, CERN/Online, Feb 2025.(Link to Indico (Access requires CERN Account))

Presentation: *Primary Vertex Reconstruction on GPU*, Tracker POG Meeting, CERN/Online, Dec 2025.(Link to Indico (Access requires CERN Account))

Awards and Affiliations

Swiss Study Foundation - Mentoring Program *Since 2024*
Selected for academic excellence and intellectual potential among Swiss university students.

YMCA Scout Leader, CEVI Altstetten-Albisrieden *Since 2016*
Led activities and security operations for 1,500+ (2022 regional camp) and 500 participants (2024 national leaders assembly).

President, FDP.The Liberals Zurich District 9 *Since 2026*
Representing the District 9 branch, VP since 2022; ran for communal (2022/2026) and cantonal (2023) elections. First in line to fill a vacancy in Zurich's City Parliament.

Projects

Modeling Charged Particle Motion in Magnetic Fields with FEM and PINNs Grade: **6/6**
Modeled magnetic fields with the finite element method in NGSolve and simulated charged particle trajectories in Python using a fourth-order Runge-Kutta scheme, then implemented and evaluated a physics-informed neural network (PINN) with PyTorch for the Lorentz force equation.

Layer-wise Analysis of Transformer Models for Speaker Identification Grade: **6/6**
Evaluated Wav2Vec 2.0, XLS-R, and Whisper for speaker identification by fine-tuning and analyzing layer-wise representations (SVCCA, k-means, t-SNE), showing different depth-dependent behavior of speaker features and identifying the optimal number of transformer layers for each model.

Skills

Programming:	Python, C, C++, C#, R
ML & Data:	PyTorch, scikit-learn, RAG, NLP, RL, Computer Vision
Scientific Computing:	FEM (NGSolve), deterministic annealing, SLURM, ROOT, CMSSW
Tools:	Git, Linux, Docker, LaTeX, Flask
Languages:	German (native), English (fluent), French (basic)